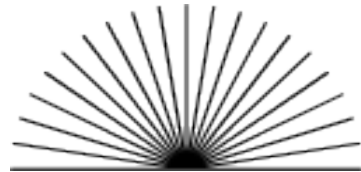


IoT · Raspberry Pi · SDS011 · ThingSpeak · MATLAB

Smart Air Quality Monitoring System

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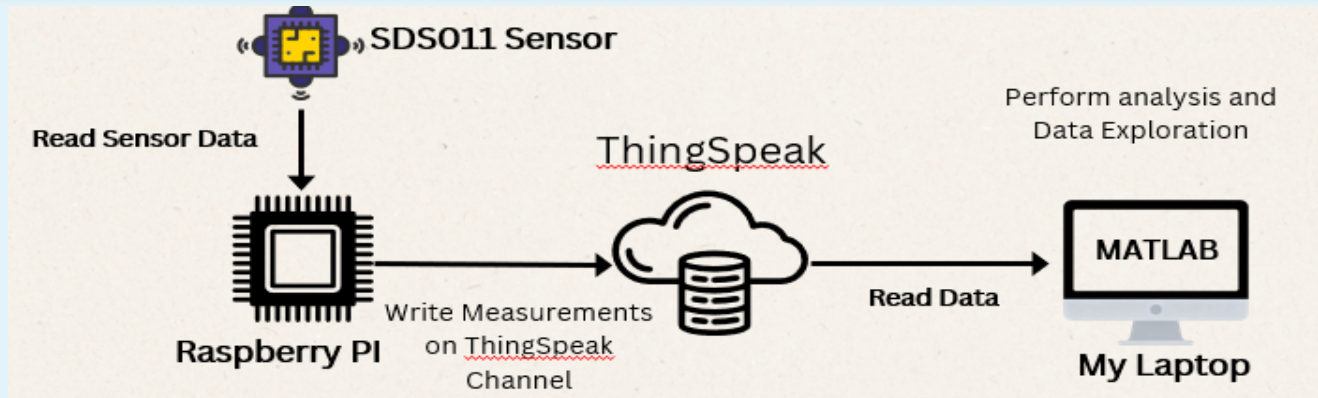
Project Overview

The Problem

Air pollution (PM2.5 & PM10) is a hidden health risk. Traditional monitoring stations are expensive, fixed, and slow, real-time local data is missing.

What We Built

A low-cost, real-time IoT air quality monitor that collects PM2.5 and PM10 data every 15 seconds, uploads to ThingSpeak cloud, and uses MATLAB for advanced analytics, anomaly detection, and WHO guideline comparison.



Cloud Integration & Live Monitoring

ThingSpeak Setup

- 1 Create free account at thingspeak.com
- 2 Make a new Channel for your sensor
- 3 Add Field 1 → PM2.5 | Field 2 → PM10
- 4 Copy Write API Key into Python script
- 5 Copy Read API Key into MATLAB script

MATLAB Data Fetch

```
channelID = YOUR_CHANNEL_ID;  
readKey   = 'YOUR_READ_API_KEY';  
data = thingSpeakRead(channelID,  
    'Fields',[1,2],  
    'NumPoints',8000,  
    'ReadKey',readKey,  
    'OutputFormat','timetable');
```

15 sec

Upload rate

Field 1

PM2.5 channel

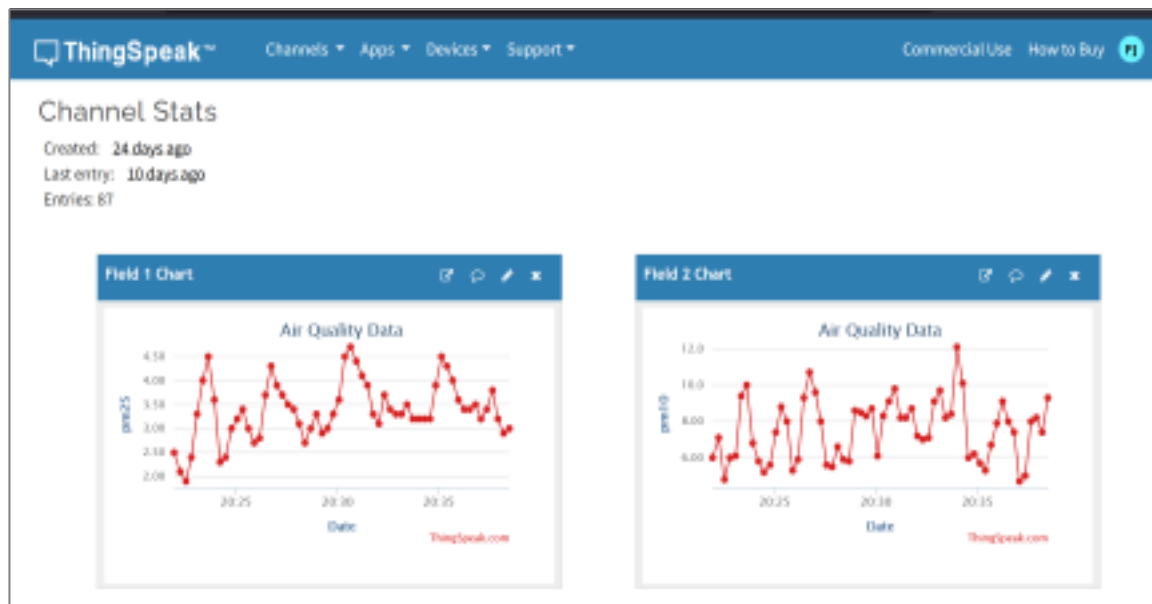
Field 2

PM10 channel

∞ Points

Stored readings

ThingSpeak Channel Analytics



87

Total Entries

15 sec

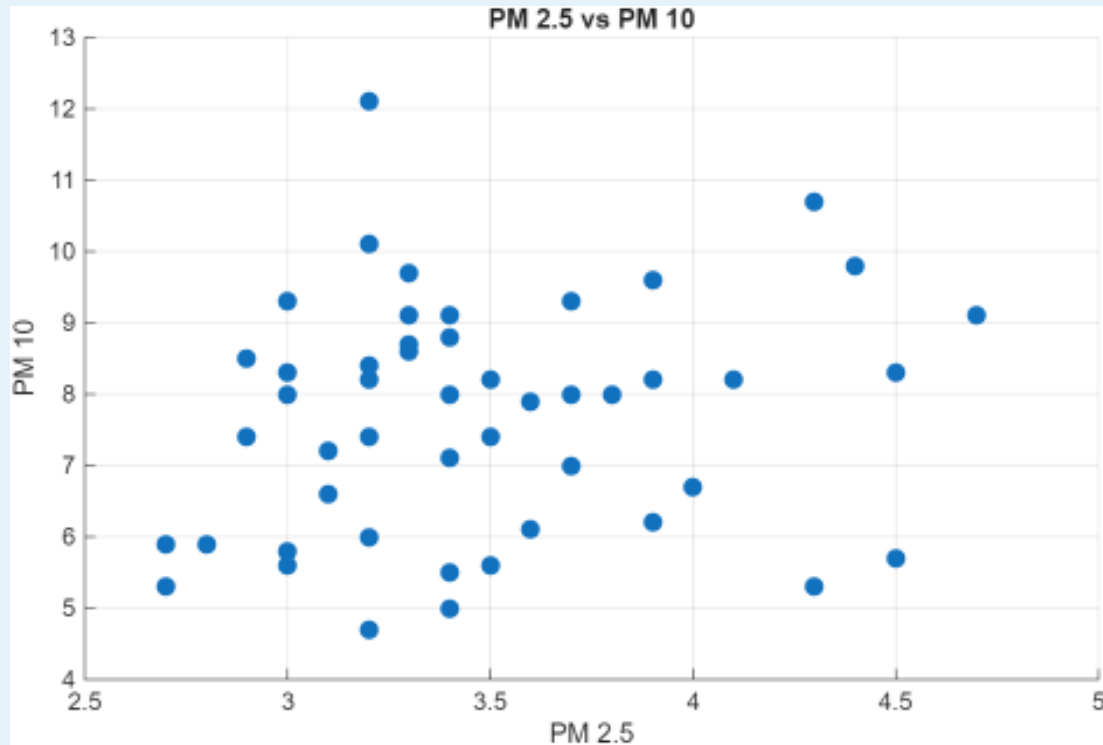
Update Rate

2 Fields

PM2.5 + PM10

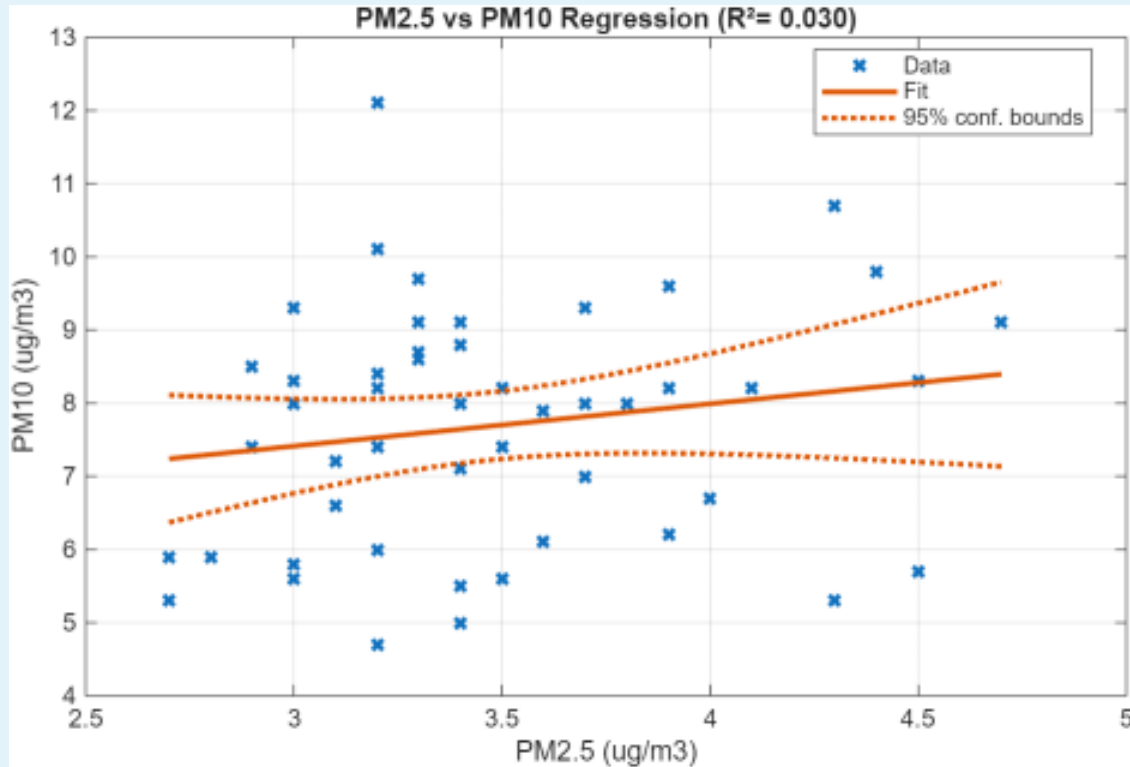
OUR ANALYSIS AND MODELLING

Correlation Analysis: PM2.5 vs PM10



Key insight: There is no distinguish relationship among these two variables (PM 2.5 and PM 10).

Fitted Regression Line among variables



RESULTS

$$R^2 = 0.03$$

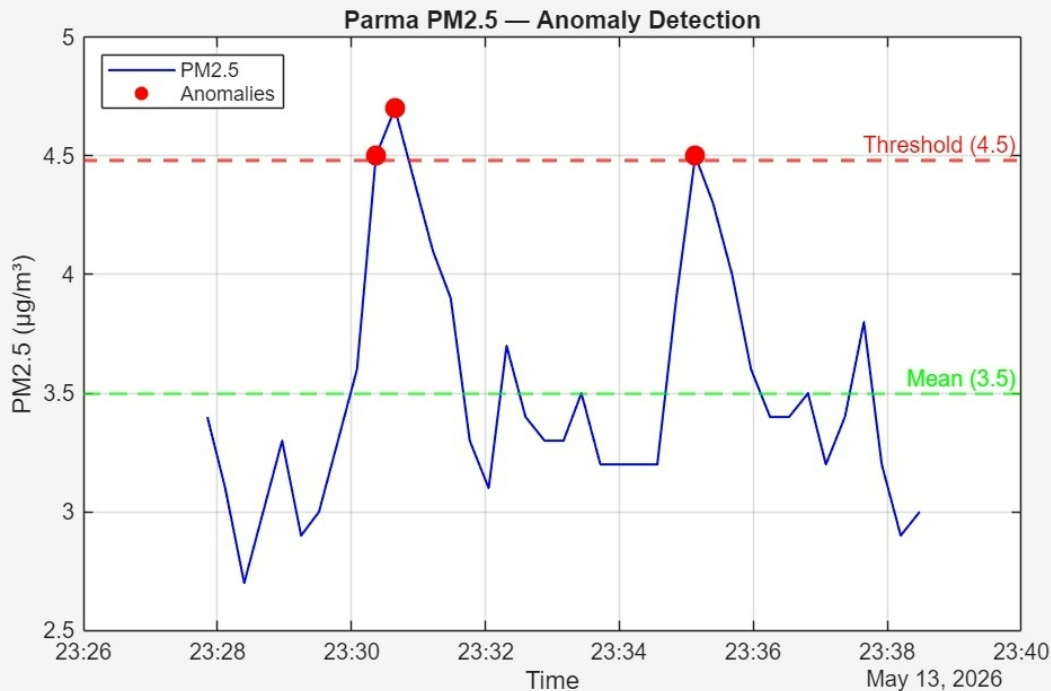
$$\text{Adjusted } R^2 = 0.0098$$

$$\text{P-value} = 0.229$$

$$\text{RMSE} = 1.63$$

Key Insight: $R^2 = 0.03$ confirms PM2.5 and PM10 come from different pollution sources. Both channels are essential for complete air quality assessment.

Anomaly Detection — PM2.5 Spike Alerts



How It Works

Calculate mean of all PM2.5 readings $\rightarrow 3.5 \mu\text{g}/\text{m}^3$

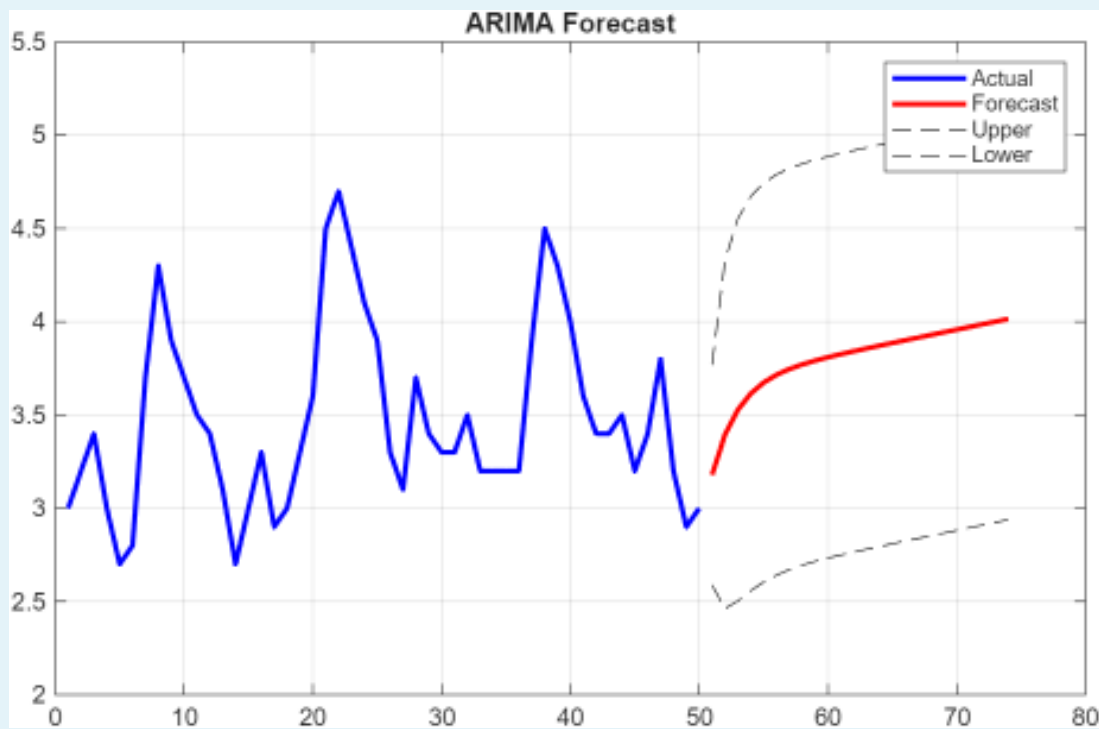
Set threshold at mean + 1 std dev $\rightarrow 4.5 \mu\text{g}/\text{m}^3$

Flag any reading above threshold as anomaly

2 anomalies found at 23:30 and 23:35 (brief pollution spikes)

Both anomalies were brief spikes — possibly from nearby traffic or combustion. Air quality returned to normal within 1-2 minutes.

ARIMA Forecasting



Insights

p=0.021 confirms yesterday's value reliably predicts tomorrow's

t-statistic of -2.53 and p-value of 0.011 make it the most statistically reliable coefficient in the model.

The historical series shows high volatility with no clear trend

Variance is highly significant

Key insight: Past errors matter more than past values

Conclusion

1

System works reliably

IoT pipeline runs 24/7 with automatic restart on Pi boot

2

Data quality is good

87 clean entries, 15-second resolution, no data gaps

3

Air quality is SAFE

Parma PM2.5 & PM10 remain well below WHO limits

4

Anomaly detection works

MATLAB flagged 2 real spikes automatically

5

Scalable architecture

Easy to expand to multiple sensors city-wide

6

Next step: ML models

LSTM forecasting when dataset grows to 30+ days

Thank you very much for your time!

